



Nonprofit Organizational Use of DEI Language: A Longitudinal Study of IRS Form 990s

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POL 688: Digital Traces
University of Arizona
April 30, 2026



Research Questions and Motivation

- RQ1: Did the use of DEI-related language increase or decrease in large U.S. nonprofit annual filings (IRS Form 990) between 2016 and 2022?
- RQ2: Does this trend vary by the category of mission focus of the organization?
- Motivated by the growing racial consciousness of the 2010s, the post-2020 racial reckoning, DEI pledges from institutional actors, and growing political backlash against DEI programs.
- An absence of longitudinal, text-based evidence makes this an open empirical question

Prior Longitudinal Work

- “Does use of emotion increase donations and volunteers for nonprofits?”
Paxton, Velasco, and Ressler (2020), American Sociological Review, 85(6)
- Used IRS Form 990 mission statement text as a corpus across 89,528 nonprofits
- Key finding: nonprofits using positive emotional language attracted more donations and volunteers
- Method: dictionary-based text analysis - same approach used in this project
- Influence on this project:
 - Established precedent for dictionary-based analysis of 990 text
 - Showed 990 language correlates with real organizational outcomes

Data: IRS Form 990, NTEE, and ProPublica

- IRS Form 990: Annual return filed by U.S. tax-exempt nonprofits (501(c)(3))
 - Required for orgs with gross receipts above \$200,000
 - Discloses mission, programs, governance, financials, and staff counts
 - One of the richest longitudinal sources of nonprofit organizational data
- NTEE Code: IRS/NCCS (Urban Institute) classification by mission area - A = Arts, B = Education, R = Civil Rights, P = Human Services
- ProPublica Nonprofit Explorer: Free database hosting 990 PDFs - the primary data source for this project
- Note: Preliminary data - scale implications discussed in Next Steps

Sample Design: Selecting Organizations

- 3 well-known, nationally recognized nonprofits per NTEE sector - 12 orgs total
- Filing consistency check: each org verified to have filed for the majority of target years
- Years: 2016, 2018, 2020, 2022 - every other year to capture pre/during/post 2020
- 48 org-year filings targeted; 44 successfully extracted; 4 unavailable via ProPublica

Arts & Culture (A): Met Museum of Art, Lincoln Center, Kennedy Center

Education (B): NAACP LDF, UNCF, Teach For America

Civil Rights (R): ACLU Foundation, National Urban League, HRC Foundation

Human Services (P): United Way, Goodwill International, American Red Cross

Data Collection: Building the Pipeline

- ProPublica does not offer bulk PDF downloads due to bot protection – for larger sample in the future I will leverage the RSelenium package in R to scrap the websites
- 990 PDFs for each of the 12 NPOs over the 4 years of interest.
- PDFs manually downloaded and saved with standardized naming: EIN_YEAR.pdf

Data Collection: Optical Character Recognition (OCR) Text Extraction

- PDFs processed using OCR via pdftools and tesseract
 - 990s are scanned image-based documents - standard PDF text extraction failed
 - OCR applied page-by-page; text consolidated into a single corpus
- Full scan of all 44 PDFs took approximately 4 hours to complete
- Output: text_corpus_index.csv - 44 org-year rows with each row containing the full extracted text of the entire IRS Form 990 PDF formatted into a single text string

Data Collection: Optical Character Recognition (OCR) Text Extraction

131624086_2016.pdf

File C:/Users/jenen/JE/2026/Arizona/Spring%202026/POL%20...

1 of 92

efile GRAPHIC print - DO NOT PROCESS As Filed Data - DLN: 93493045015347

Form 990

Department of the Treasury Internal Revenue Service

Return of Organization Exempt From Income Tax

OMB No 1545-0047

Under section 501(c), 527, or 4947(a)(1) of the Internal Revenue Code (except private foundations)

Do not enter social security numbers on this form as it may be made public open to public treasury information about form 990 and its instructions is at www.irs.gov/form990 . inspection internal revenue service a for the 2015 calendar year, or tax year beginning 07-01-2015 , and ending 06-30-2016

A For the 2015 calendar year, or tax year beginning 07-01-2015 , and ending 06-30-2016

B Check if applicable

☐ Address change

☐ Name change

☐ Initial return

☐ Final return/terminated

☐ Amended return

☐ Application pending

C Name of organization

METROPOLITAN MUSEUM OF ART

% CONTROLLER'S OFFICE

Doing business as

Number and street (or P O box if mail is not delivered to street address) Room/suite

1000 FIFTH AVENUE

City or town, state or province, country, and ZIP or foreign postal code

new york, ny 100280198

g gross receipts \$ 1,238,769,849

f name and address of principal officer h(a) is this a group return for thomas campbell > 1000 fifth avenue su pordinates [yes iv new york,ny 100280198 h(b) are all subordinates [yes [no t tax-exempt status ff so4cy3) [7 sot(c)() @(insertno) [7 4947(a)(1) or [7 527 included? if "no," attach a list (see instructions) j» website: » www.metmuseum.org h(c) group exemption number > k form of organization [¥

Ln 1, Col 1 191,657 characters 100% Windows (CRLF) UTF-8

Constructing the DEI Lexicon: Sources

- DEI language is not self-defining - a systematic word list is needed
- Cornell University DEI Glossary: Academic and HR terminology
 - e.g., equity, microaggressions, systemic racism, implicit bias, belonging
- NSF Leaked DEI Trigger Word List (2025): Government-flagged terms
 - e.g., underrepresented, diversify, barriers, inequalities, equal opportunity
- Both source URLs reviewed and extracted with AI assistance (Claude)
- Combined: aspirational DEI language and politically contested terms

Constructing the DEI Lexicon: Cleaning

- Initial lexicon: approximately 130 unique terms
- 6 terms removed as likely false positives:
 - Ability - overwhelmingly used as ability to
 - Respect - overwhelmingly used as with respect to
 - Liberation - arts sector; refers to artistic programming, not DEI
 - Excluded - financial usage: excluded from gross revenue
 - Historically - temporal: historically, the organization has...
 - Culture - Arts sector artifact; refers to artistic culture
- Final cleaned lexicon used in all scoring

Constructing the DEI Lexicon: Cleaning

acknowledgement	activism	diversity	transparency
dialogue	barrier	inclusion	multicultural
equity	barriers	leadership	victim
cultural heritage	disabilities	racism	equality
discriminatory	equitable	solidarity	gender
privileges	marginalize	advocate	latinx
impact	promoting	diversify	underserved
race	underrepresented	women	people of color
fostering	diversified	empathy	binary
justice	advocacy	inclusive	discrimination
social justice	racial	indigenous	lgbtq
resilience	gender identity	marginalized	bias
sexual orientation	black and latinx	bipoc	activists
			growth mindset

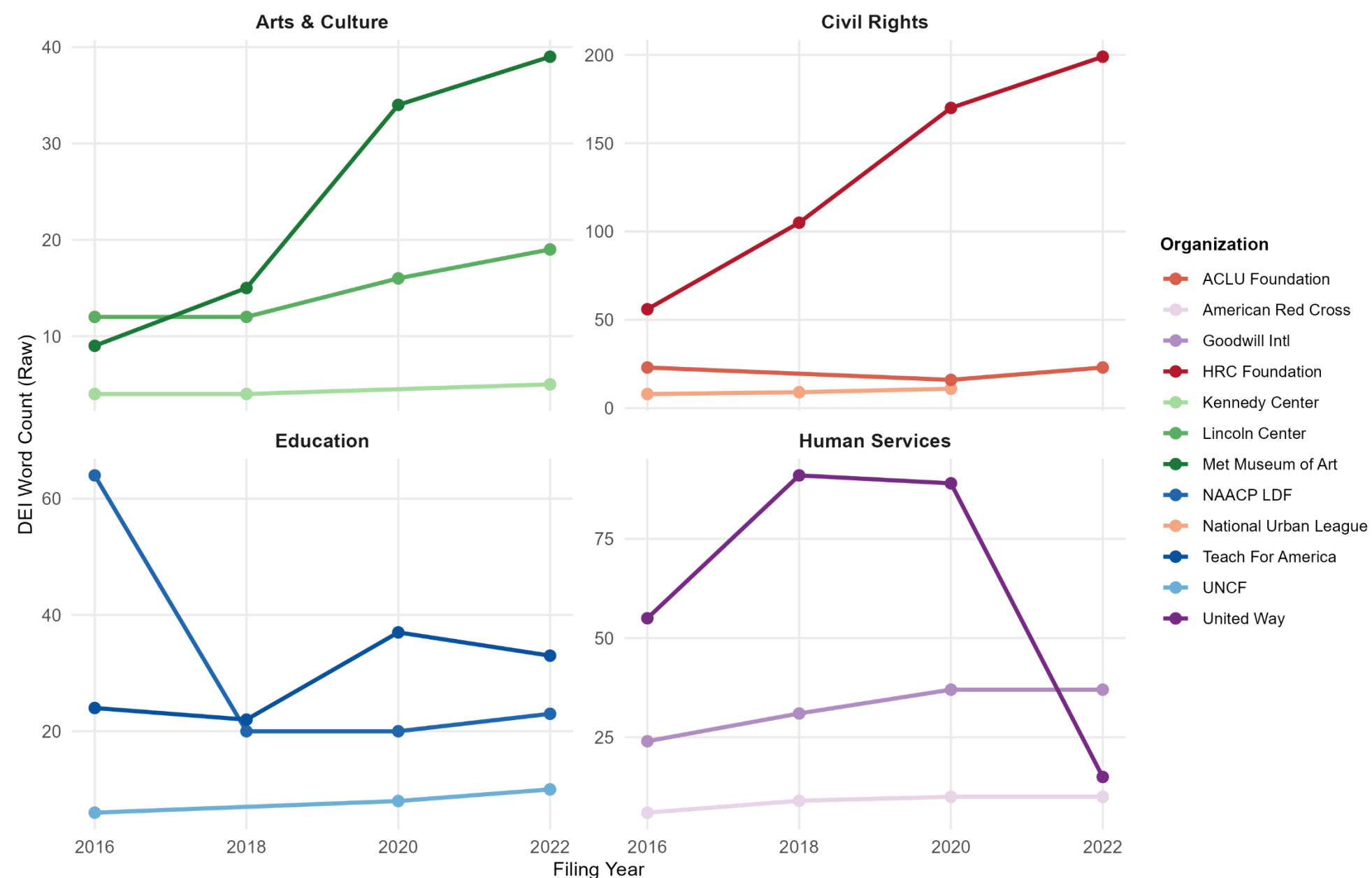
Scoring: Dictionary-Based Text Analysis

- Dictionary/frequency matching
 - Each 990 lowercased; terms counted via exact string match (`str_count()`)
 - Output: total org-year DEI word count and term-level breakdown per document (total usage of specific term)
- Outputs: `dei_scores.csv` (org-year) and `dei_term_counts.csv` (term-level)
 - 53 of the 124 words in the DEI Lexicon used
- Why dictionary over machine learning?
 - Transparent, reproducible, and interpretable
 - Consistent with Paxton et al. (2020) using 990 mission statement text
 - Compromise on nuanced understanding of DEI terms.

Results: DEI Language Trends by Organization and Sector (2016-2022)

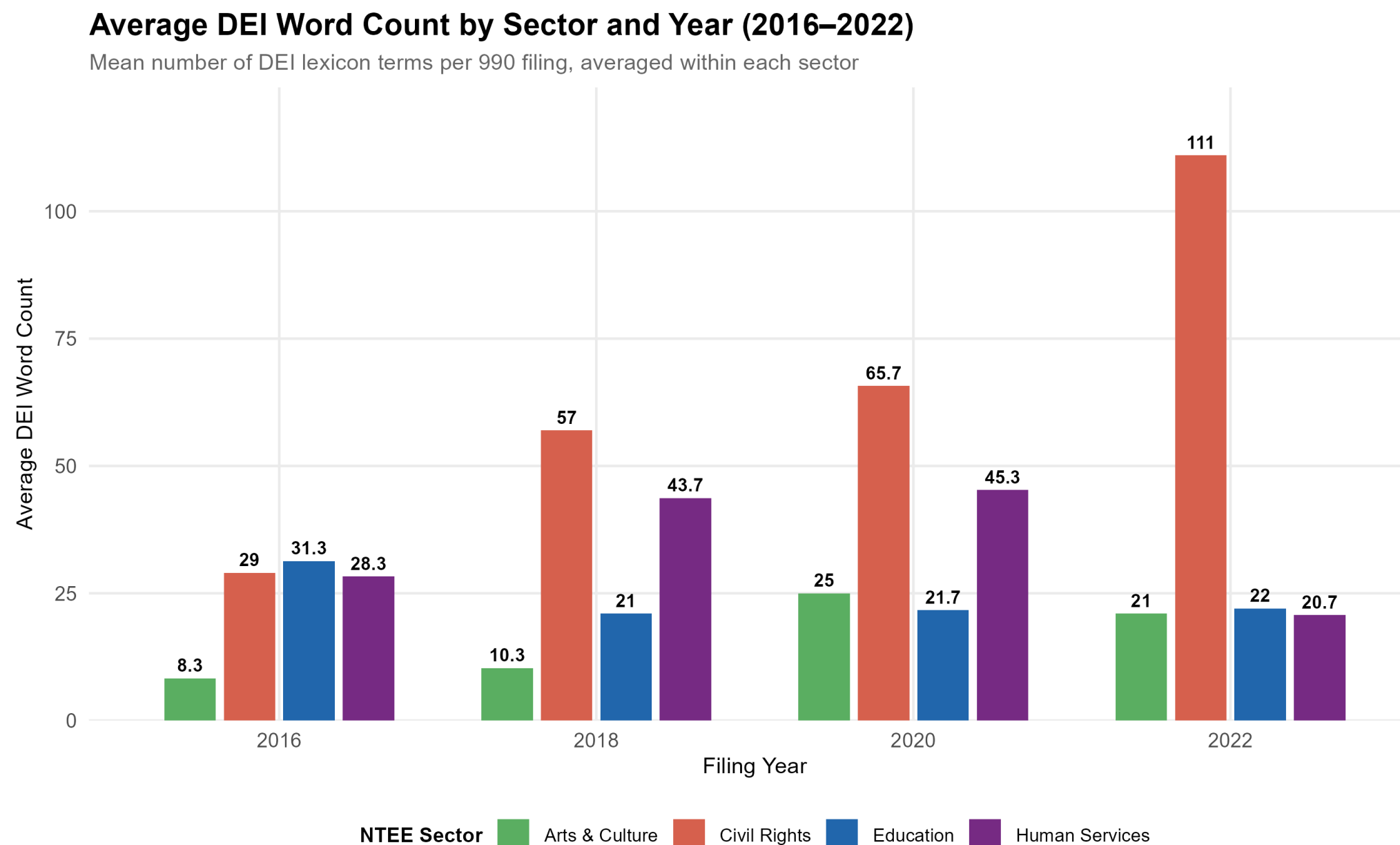
DEI Language in Nonprofit 990 Filings by Sector (2016–2022)

Total count of DEI lexicon terms per filing year



Sources: IRS Form 990 filings via ProPublica. DEI lexicon from Cornell University DEI Glossary and NSF leaked trigger word list (2025). Note: some org-years missing due to unavailable PDFs.

Results: Average DEI Language by Sector Over Time (2016-2022)



Sources: IRS Form 990 filings via ProPublica. DEI lexicon from Cornell University DEI Glossary and NSF leaked trigger word list (2025).

Descriptive Summary

- DEI word counts ranged from single digits (Kennedy Center, 2016) to nearly 200 (HRC Foundation, 2022)
- The 2020 filing year shows the largest average increase of DEI language use across all sectors
- Sector matters: Civil Rights orgs use 3 to 4 times more DEI language than Arts and Culture orgs
- No org showed a consistent decline across all four years - the overall trajectory is upward

Challenges: Data Collection

- Manual PDF download: No bulk download from ProPublica - does not scale
 - RSelenium package to bypass bot protection on each individual URL
- Missing filings: 4 org-year PDFs unavailable - limits panel balance
- PDF format variability: Low-resolution scans affect OCR accuracy
- OCR runtime: Scanning 44 PDFs took approximately 4 hours locally

Challenges: Text Extraction and Measurement

- OCR noise: Garbled characters, split words, missed text on low-quality scans
- Lexicon validity: Dictionary methods cannot detect negation or context
- No normalization: Raw counts favor longer 990s regardless of actual DEI emphasis.
 - Could instead take DEI count as a proportion of all text in 990.
- Small N: 44 documents is descriptive, not inferential - findings cannot be generalized

Tools

- R - primary analysis and visualization language
- pdftools and tesseract - PDF processing and OCR text extraction
- tidyverse and stringr - data wrangling and dictionary-based term counting
- Ggplot2 - all three visualizations
- ProPublica Nonprofit Explorer - primary data source for 990 PDFs
- Claude (AI) - DEI term extraction

Next Steps: Scaling the Sample

- Current study: 12 orgs, 4 NTEE sectors, 4 years - a proof of concept
- Expand organizations: Scale to thousands of organizations, including mid-size and smaller orgs
- Expand NTEE coverage: Add Health (E), Environment (C), Religion (X), Youth Development (O), and others
- Expand year coverage: Annual filing data from 2010 through 2024
- Automate data collection: R-based ProPublica API script replaces manual downloads

Next Steps: Computational Infrastructure (HPC)

- Current pipeline runs locally - feasible at 44 documents, not at thousands
- University of Arizona HPC cluster will be used for the scaled version:
 - Parallel OCR - 4 hours locally becomes minutes on the HPC
 - Dictionary scoring across thousands of org-year documents simultaneously
 - Topic modeling (LDA) to identify DEI sub-themes at scale
 - Determine if the sub-themes of DEI language correlate with year of NTEE

Conclusion

- This project developed a reproducible, scalable pipeline for measuring DEI rhetoric in IRS Form 990 filings
- Preliminary finding:
 - DEI language use between 2016 and 2022 varied amongst NTEE
 - There was a growth in Civil Rights NPO usage (especially after 2020)
 - A decrease in usage for Human Service NPOs especially after 2020
 - Modest fluctuation in usage for Education and Arts and Culture NPOs.